CONTEXT OF THE BUSINESS

Turtle Games is a company that both manufactures and sells its own product, while also sourcing and retailing products made by the other company. Its products are books, board games, video games, and toys. I am part of the team of data analysts whose purpose is improving overall sales performance by analysing data and customer review collected by the Turtle Games. The analysis I will provide will try to answer some question posed by Turtle Games:

1. What does influence the loyalty point?
2. How can we cluster the customers for marketing purpose?
3. How can we leverage social data once customer reviews?
4. Can prescriptive statistics demonstrate the suitability of the loyalty points data to create predictive models?

ANALYTICAL APPROACH

The dataset consist in 2000 observations and 11 variables. I decided to conduct my analysis using Python and RStudio. I began with Python by importing the data frame provided by Turtle Games. I focused my attention on 'loyalty point', 'age', 'remuneration', and 'spending score' columns. The data were clean, no null values were found.

The first step was to import libraries:
- numpy and pandas to work with arrays
- statsmodels to estimate statistical models and perform statistical test
- matplotlib to create visualisation
- sklearn for statistical modelling

I chose sklearn.model_selection model in order to create three linear regression which can put in a relation 'loyalty point' with the other three variables. The second step was to import mean_squared_error from sklearn to predict on training and testing sets using the decision three regressor. The first model was overfitting the train set, so there was a risk that it might not generalise to other new data in the future. I decided to apply some pruning to the model and the result was still a good fitting avoiding the problem of overfitting.

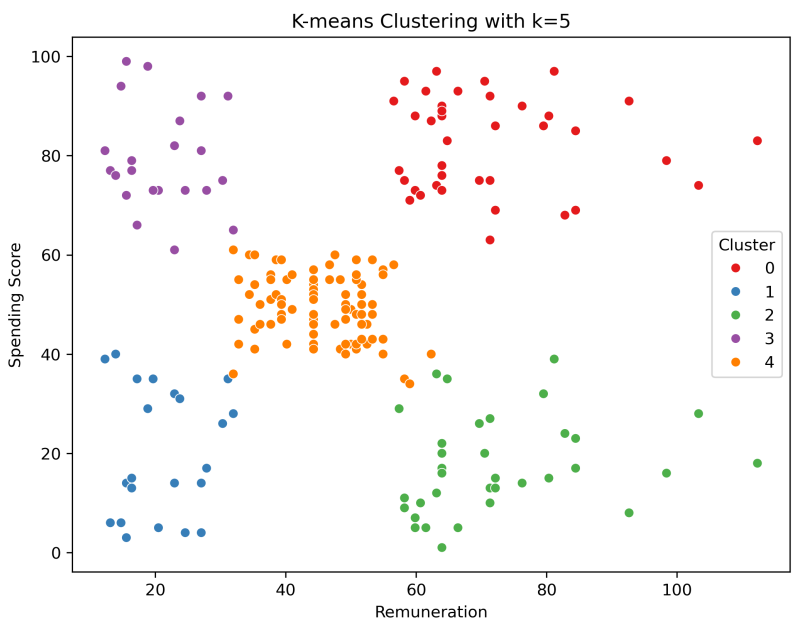
The third step was to import seaborn and start the clustering process. I used re and WordCloud libraries for the sentiment analysis. I did some attempts looking for the best fit. Regarding the sentimental analysis the first step was to delete duplicate values. I plot the most frequent words and after tokenise the 'reviews' and 'summary' columns I decide to plot the polarity of both on histograms. I finally chose to analyse the top 20 positive and the top 20 negative of both reviews and summaries.

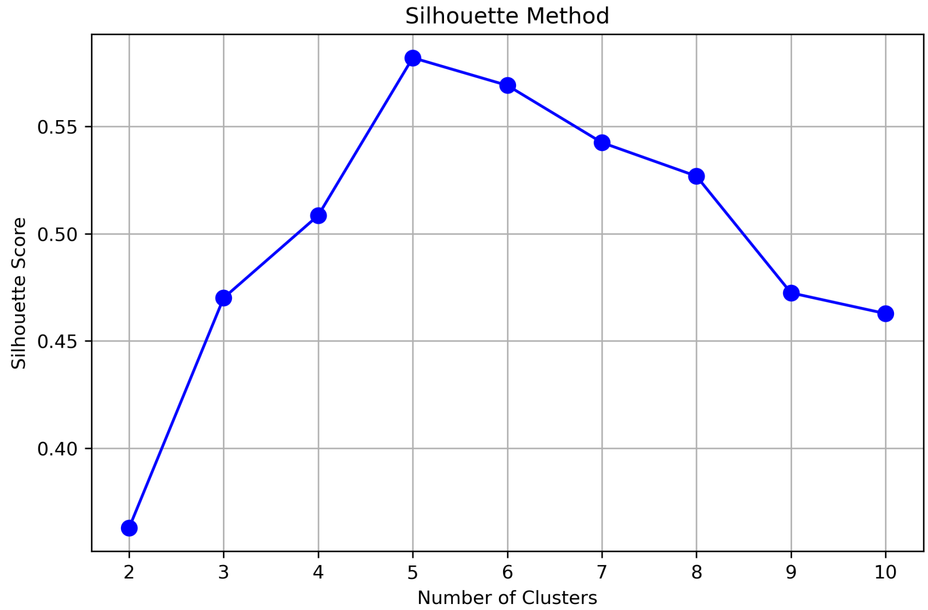
In the analysis I conduct ed with RStudio I used library as skimr to summarise the datasets, ggplot for data visualisation. After overviewing the dataset I decided to work with ‘age’, ‘remuneration’, ‘spending score’, ‘gender’, ‘education’ and of course ‘loyalty point'. I decided to create a visual distribution in histograms, boxplot and scatterplot. I later did a descriptive statistics for confirmation of the visual analysis and to gain more knowledge regarding the distribution of the data. Firstly I built a multiple linear regression with only spending score and remuneration because I deduct from the statistical report and from the single linear regression that ‘age’ wasn’t so valuable. But in a multi linear regression there is a slight improvement so I decided to integrate ‘age’ variable. I later checked for normal distribution plotting the residual in order to validate the normality assumption of the model.

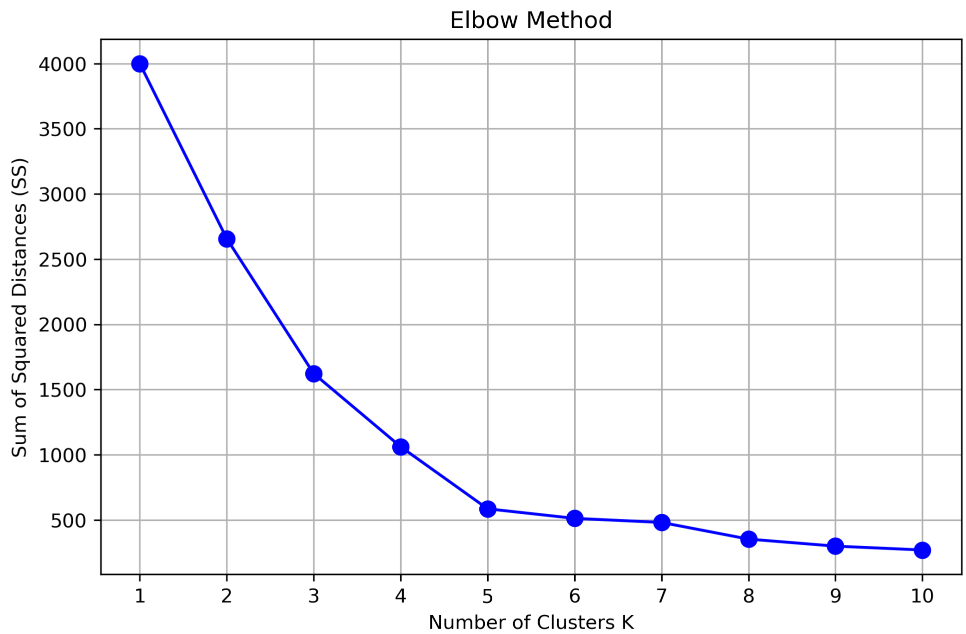
VISUALISATION AND INSIGHTS

In order to answer to the goals of this analysis I chose some visualisation that highlight the distribution of the variables, and can be useful for marketing purpose.

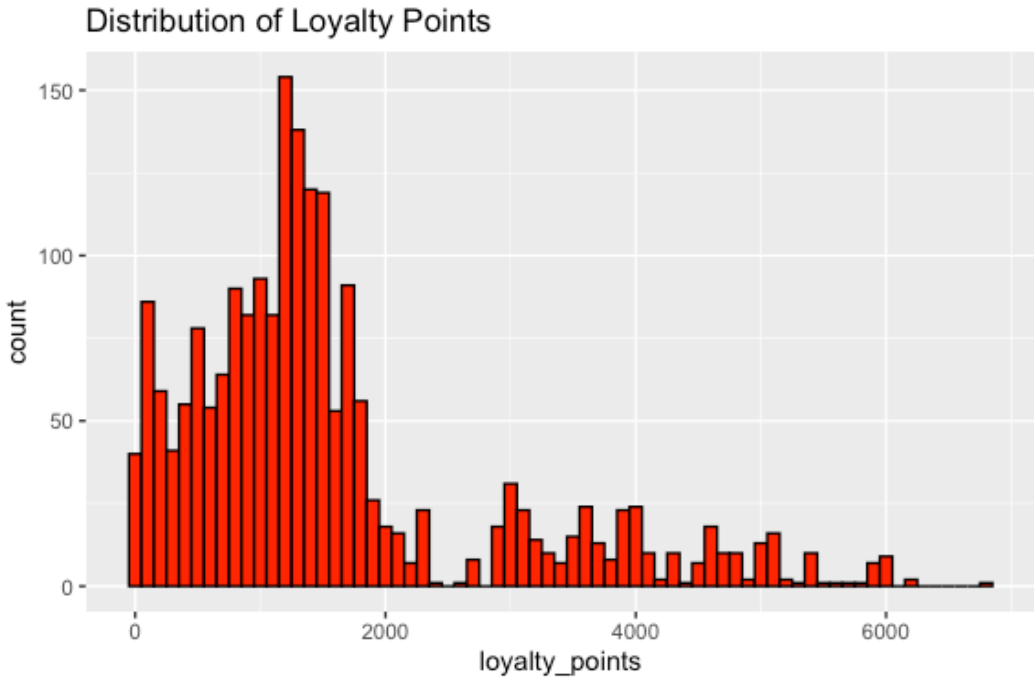
For clustering propose I used silhouette methods, elbow method, histograms, and scatterplot methods. The first two methods helped selecting the 5 cluster in a clear customers segmentation. This analysis was crucial in clustering the customers from high-income high-spending to low-income low-spending providing the marketing department instruments for targeted personalised incentives. It is clear form the three graphics that 5 is the best fitting cluster number.

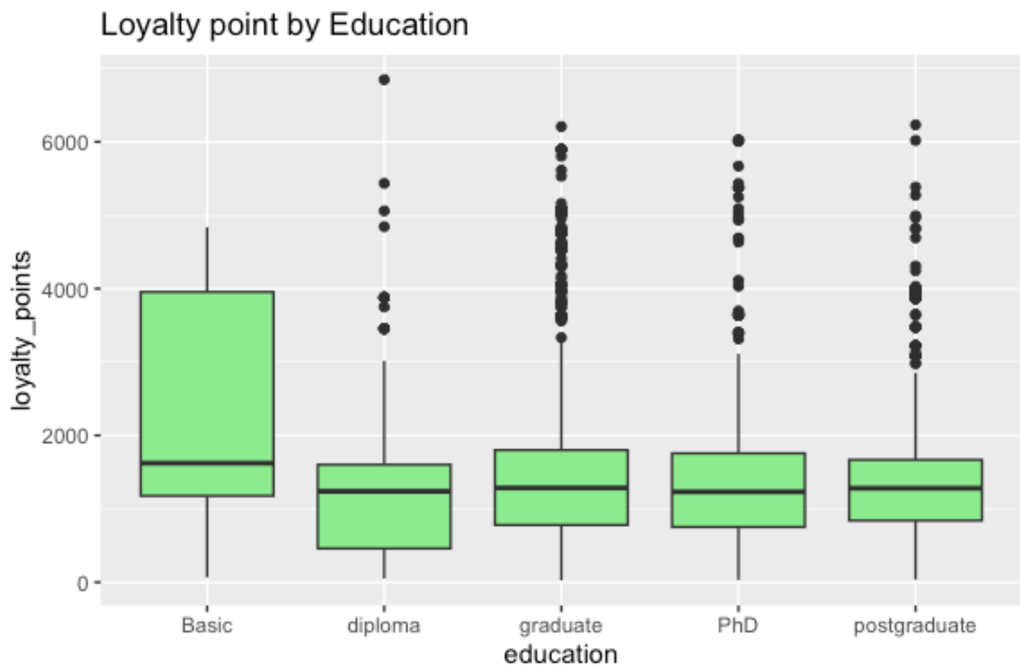




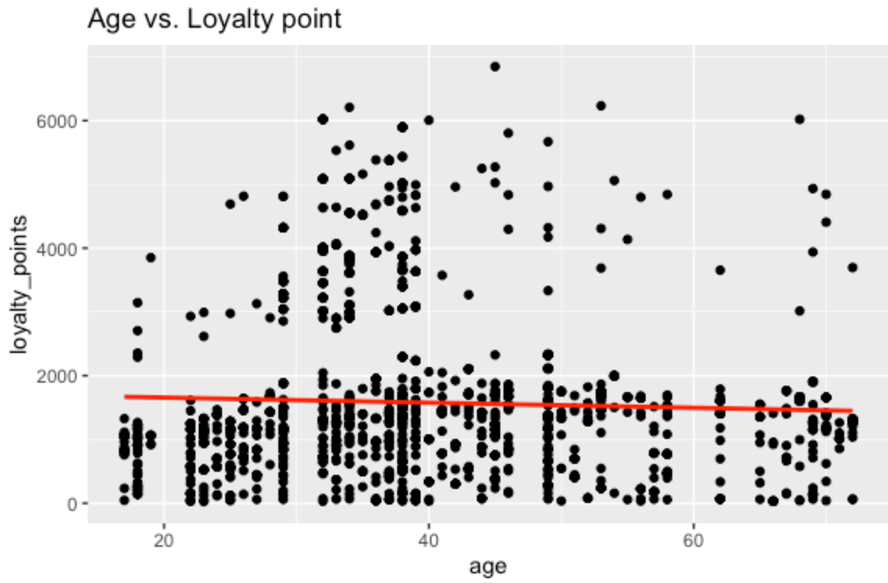


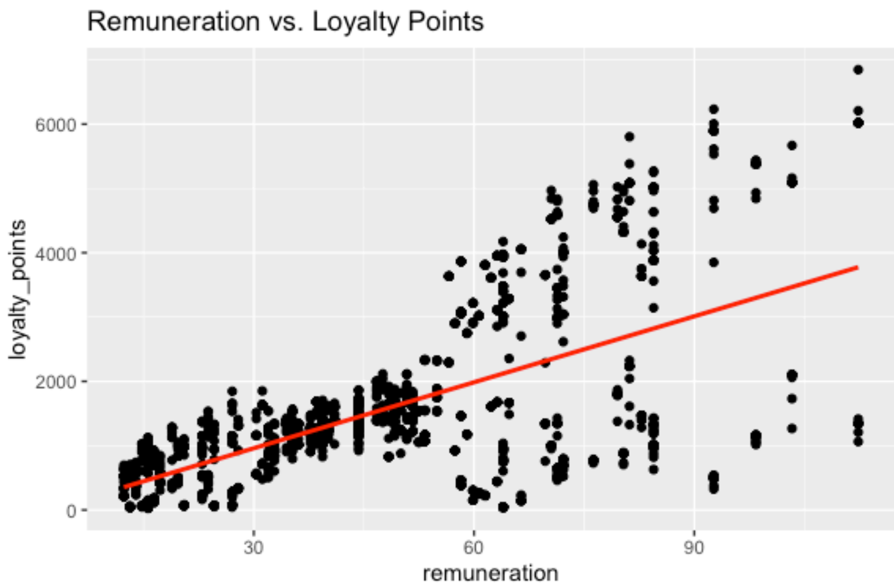
I chose histogram and box plots to investigate the distributions of the variables. The loyalty points histogram showed a right-skewed distribution, where the majority of customers accumulate relatively low points, while a few outliers earn a vast amount. This pattern is shown even in the box plot where we can see that the outliers are mostly well-educated.

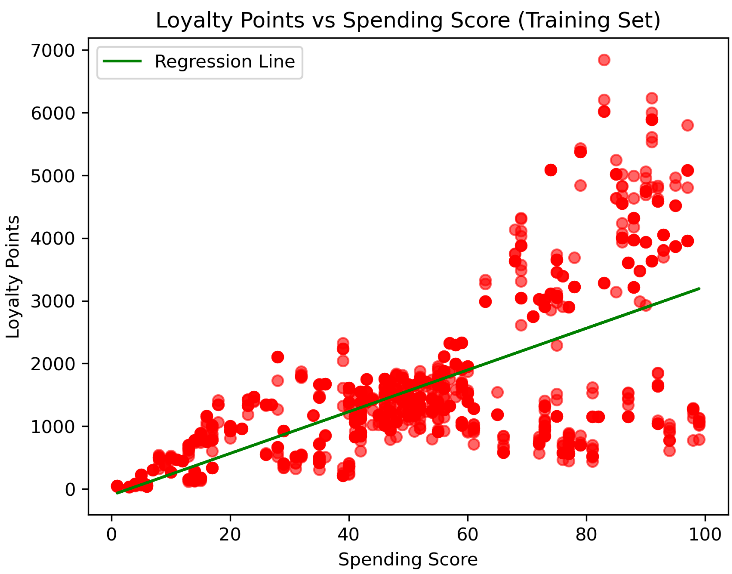




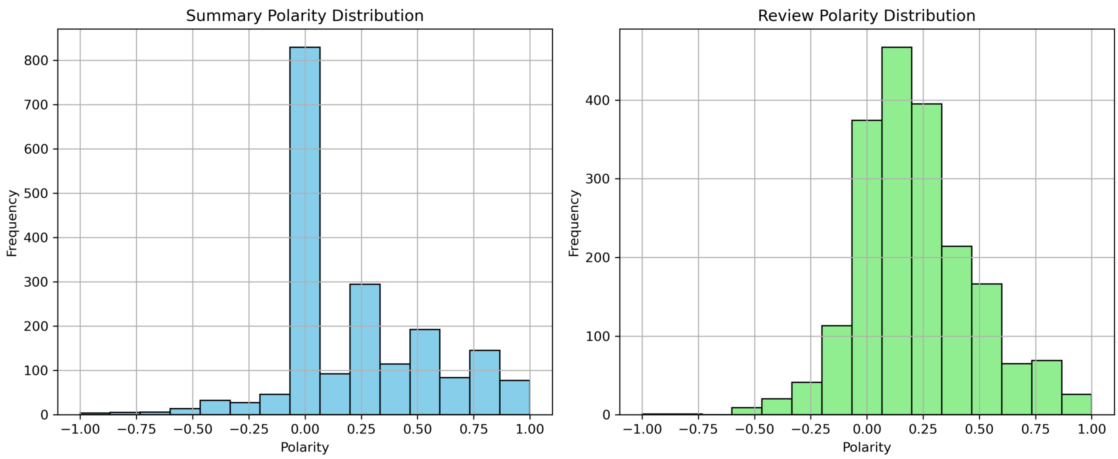
To deepen the analysis, I used scatterplot and a regression line to visualise the relationship between these three numerical variable. I detected a positive correlation among loyalty point and remuneration(0.62), and among loyalty point and spending score (0.67). While there was a visually undetectable correlation among age and loyalty point (-0.04). Surprisingly there is a slightly negative correlation among age and spending score.


Age vs. Loyalty point
loyalty_points
6000
4000
2000
0
20
40
60
age

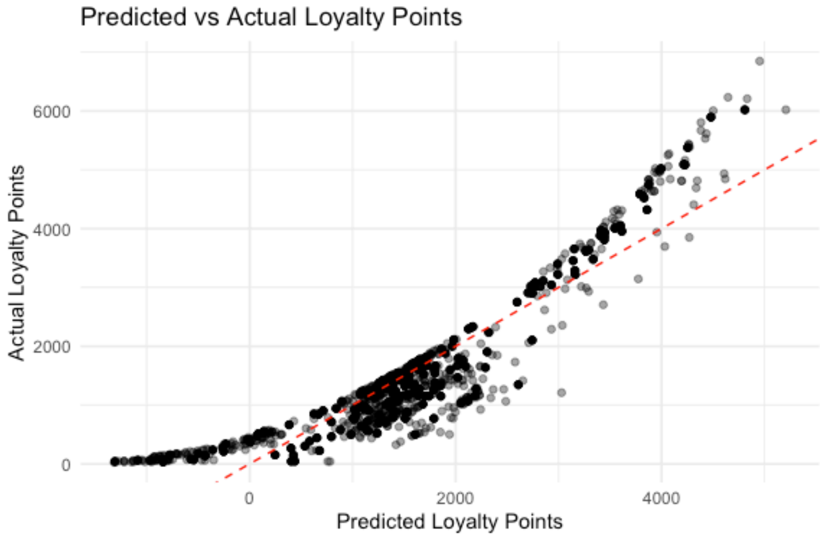

Remuneration vs. Loyalty Points
loyalty_points
6000
4000
2000
0
30
60
90
remuneration

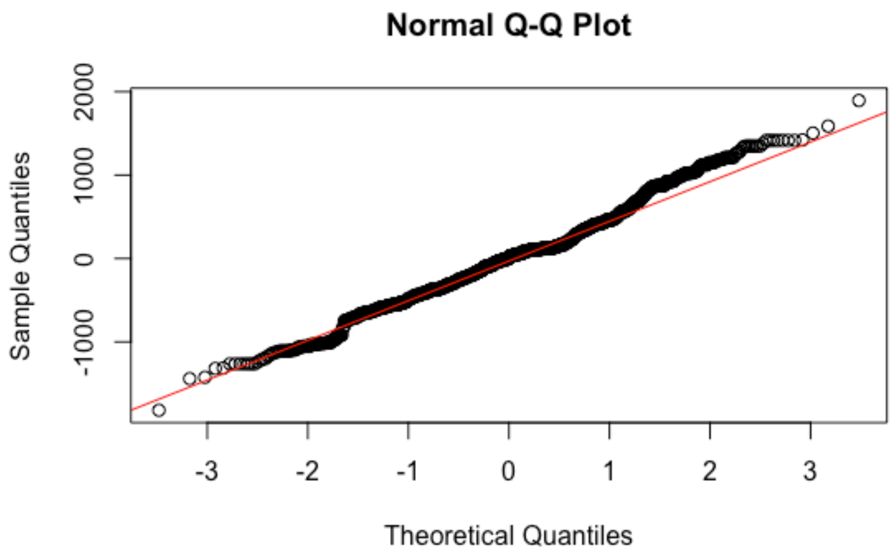

Loyalty Points vs Spending Score (Training Set)
Regression Line
Loyalty Points
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
Spending Score

Regarding the qualitative insights in the sentiment analysis, I used words clouds and polarity histograms. The first provided a quick glance at words as ‘stars’, ‘five’, and ‘great’. While the polarity histograms revealed a more positive than negative sentiments for both ‘summary’ and ‘reviews’ variables. As we can see from the histograms there is a left-skewed distribution which shows a more positive than negative sentiment.



The multi linear regression model built can predict 83.99% of the variance. This model can support marketing decision, clustering customers in 4/5 identify customers and check who is underperforming and create incentives suited for target customers. In the second graph below I check the normality of the residuals in order to validate the assumptions of the model and as shown the residuals are generally distributed around the red line, which could exclude the log-linear attempt for the creation of a more fitter new model.





PATTERNS AND PREDICTIONS

The analysis revealed some interesting insights. Loyalty point is strongly and positively influenced by spending score and remuneration, indicating that the wealthier customers spend more money, so they tend to accumulate more spending score and they get rewarded by more loyalty points. Even if age has a weak correlation with loyalty point its inclusion in the multiple linear regression model has a slight improvement. The decision tree we built have a strong fit and can classify new data 91% of the time.

The multi linear model can predict 83.99% of the variance. This model can support marketing decision, clustering customers in five groups and create incentives suited for them:

- high both remuneration and spending
- low both remuneration and spending
- high remuneration, low spending
- low remuneration, high spending
- medium both remuneration and spending

There is frequency toward a more positive sentiment than negative sentiment, indicating that more comments express satisfaction than disappointment. So the 'brand reputation' remains strong. The negative comments in both summary and reviews ('boring' and 'disappointing' are very frequent words) reveal some key area for improvements.